



F21 - Beregnelighed og logik [520171U005]



Tests, Surveys and Pools Tests

Test Canvas : RE-EXAM C&L 13 August 2021

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Description This is the official re-exam Computability & Logic 2021

Instructions This exam consists of 40 questions (in random order)

You have 3 hours to finish the exam (unless you got dispensation)

You can only submit this exam once. You cannot resubmit it.

When the timer expires, the exam is submitted automatically.

In case of urgent technical problems, mail me at jaco@cs.au.dk or call me on 44 79 19 71 00

Total Questions 40

Total Points 400

Select: All None Select by Type: - Question Type - ▼

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Points: **10**

1. Multiple Choice: Cardinality (var): Which one of the following propositions...

...

Question Which one of the following propositions on cardinality is CORRECT?

Answer $\mathbb{R}$ is countable

$\mathbb{Q}$ is not countable

$\mathbb{N}$ is not countable

✓ The set of triples $\mathbb{N} \times \mathbb{N} \times \mathbb{N}$ is countable



Points: **10**

2. Multiple Answer: CFL (var): Which of the following languages over...

ooo

Question	Which of the following languages over the alphabet $\{a, b, c\}$ is context-free? (you must select all context-free languages)
Answer	✓ $\{ ww' \mid n_a(w) = n_c(w') \}$ $\{ a^i b^k c^k \mid i > k \text{ and } k > 0 \}$ ✓ $\{ a^i b^k c^m \mid k > 0 \text{ and } i > m > 0 \}$ $\{ a^i b^i c^m \mid i > m > 0 \}$



Points: **10**

3. Multiple Choice: Chomsky Normal Form: Consider the following Context-Free G...

ooo

Question

Consider the following Context-Free Grammar G.

$$\left\{ \begin{array}{lcl} S & \rightarrow & DC \\ A & \rightarrow & B \mid CD \\ B & \rightarrow & \Lambda \mid DA \\ C & \rightarrow & BB \mid c \\ D & \rightarrow & a \end{array} \right.$$

We transform G into an equivalent grammar in Chomsky Normal Form.
What is the result?

Answer

$$\left\{ \begin{array}{lcl} S & \rightarrow & DC \\ A & \rightarrow & CD \mid DA \mid a \\ B & \rightarrow & DA \mid a \\ C & \rightarrow & BB \mid c \\ D & \rightarrow & a \end{array} \right.$$

$$\left\{ \begin{array}{lcl} S & \rightarrow & DC \\ A & \rightarrow & CD \\ B & \rightarrow & DA \\ C & \rightarrow & BB \mid c \\ D & \rightarrow & a \end{array} \right.$$

$$\left\{ \begin{array}{lcl} S & \rightarrow & DC \mid D \\ A & \rightarrow & B \mid CD \mid D \\ B & \rightarrow & DA \mid D \\ C & \rightarrow & BB \mid B \mid c \\ D & \rightarrow & a \end{array} \right.$$



$$\left\{ \begin{array}{lcl} S & \rightarrow & DC \mid a \\ A & \rightarrow & CD \mid DA \mid a \\ B & \rightarrow & DA \mid a \\ C & \rightarrow & BB \mid DA \mid c \mid a \\ D & \rightarrow & a \end{array} \right.$$

Points: **10****4. Multiple Answer: Closure Properties CFL (var): Given two regular languages R1 and R2...**

ooo

Question	Given two regular languages R_1 and R_2 , and two context-free languages C_1 and C_2 , which of the following languages MIGHT NOT be context-free?
Answer	$(R_1 \cup R_2) \cap C_1$ $(R_1 \cap R_2) \cup C_1$ ☒ $(R_1 \cup C_1) \cap C_2$ $(R_1 \cap C_1) \cup C_2$

Points: **10****5. Multiple Choice: CYK matrix (var): We consider the CYK-parsing algorithm...**

ooo

Question

We consider the CYK-parsing algorithm for the following grammar, which is already in Chomsky Normal Form. We use the alphabet $\{a,b,c\}$ and the non-terminals $\{S,X,A,B\}$.

$$S \rightarrow AX \mid c$$

$$X \rightarrow SB$$

$$A \rightarrow a \mid BA$$

$$B \rightarrow b$$

We run the CYK-parsing algorithm on the input word **bbacb**. We have already partially filled the CYK-matrix, but left some question marks open, as follows:

B	?	?	?	S
—	B	?	?	?
—	—	A	?	?
—	—	—	S	?
—	—	—	—	B

The question is: How many A's appear in this matrix in total after the CYK algorithm is finished?

Answer

1

2

☒ 3

4



6. Multiple Choice: Hoare triple (var): Which of the following Hoare triples ...

ooo

Question

Points: **10**

Which of the following Hoare triples is valid?	
Answer	$\{y = z\} x := y + z; z := x + y \{y = z\}$ ☒ $\{y + z = 0\} x := y + z; z := x + y \{y = z\}$ $\{x = 0\} x := y + z; z := x + y \{y = z\}$ $\{True\} x := y + z; z := x + y \{y = z\}$

Points: **10**

7. Multiple Choice: Hoare Logic invariant: Consider the following Hoare triple, ...

ooo

Question	Consider the following Hoare triple, where we write $\{\{P\}\} S \{\{Q\}\}$ for the Hoare triple for program S with precondition P, postcondition Q $\{\{x = 0 \wedge z \geq 0\}\}$ $y := 0$ $\text{while } (x < z) \{$ $x := x + 1;$ $y := y + x$ $\}$ $\{\{2y = z * (z + 1)\}\}$ We want to prove the partial correctness of this program, using the partial-while rule. Which formula below can we use as the invariant I?
Answer	☒ $2y = x * (x + 1) \wedge x \leq z$ $2y = x * (x + 1)$ $2y = x * (x + 1) \wedge x = z$ $2y = z * (z + 1) \wedge x \leq z$

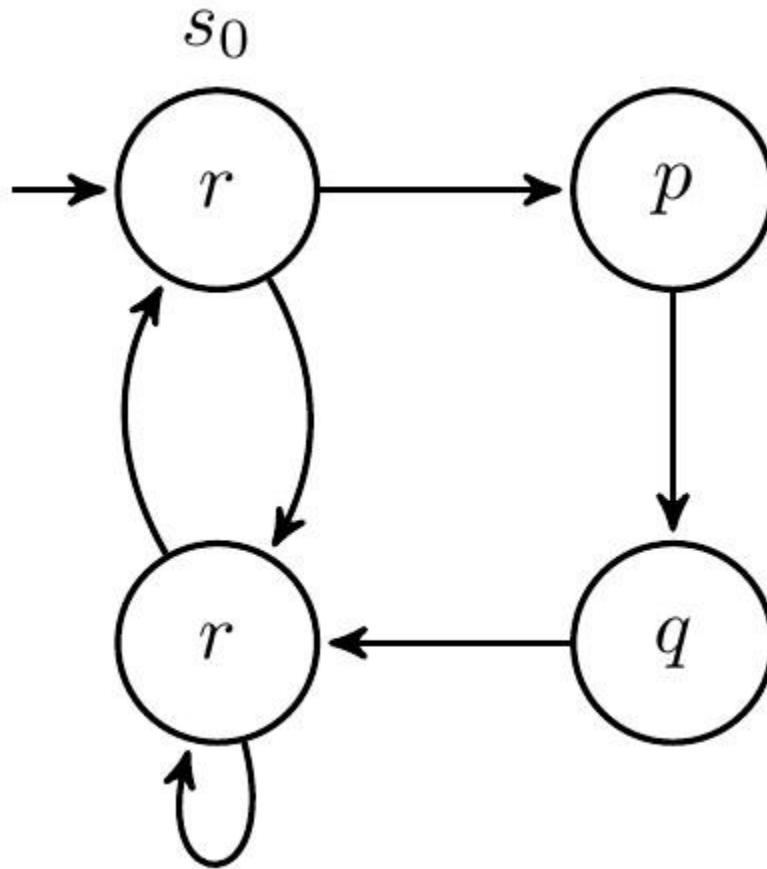
Points: **10**

8. Multiple Answer: LTL (var): Which of the LTL formulas below hold ...

ooo

Question

Which of the LTL formulas below hold for all paths from the initial state s_0 in the following Kripke structure?



(you must select all correct answers)

Answer

$F G \neg q$

✓ $G F r$

$(p \vee r \vee q) U q$

✓ $(\neg q) W q$



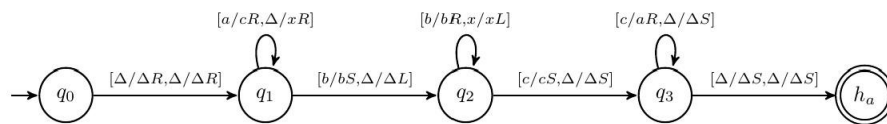
9. Multiple Choice: Multi-tape TM (2): What language is accepted by the foll...

ooo

Points: **10**

Question

What language is accepted by the following multi-tape TM with two tapes?

**Answer**

$\{a^m b^n c^m \mid m > 0 \text{ and } n > 0\}$

✓ $\{a^m b^m c^n \mid m > 0 \text{ and } n > 0\}$

$\{c^m b^n a^m \mid m > 0 \text{ and } n > 0\}$

$\{c^m b^m a^n \mid m > 0 \text{ and } n > 0\}$

$\{a^m b^m c^m \mid m > 0\}$

$\{c^m b^m a^m \mid m > 0\}$



Points: **10**

10. Multiple Answer: Natural Deduction (predicate): Below you find four derivations in na...

ooo

Question

Below you find four derivations in natural deduction for predicate logic. However, two derivations are correct, and two derivations are wrong. Select the two CORRECT derivations.

Answer

1. $\forall x \exists y P(x, y)$ premise
2. x_0 fresh variable
3. $\exists y P(x_0, y)$ $\forall e, 1$
4. y_0 fresh variable
5. $P(x_0, y_0)$ assumption
6. $\forall x P(x, y_0)$ $\forall i, 5$
7. $\exists y \forall x P(x, y)$ $\exists i, 6$
8. $\exists y \forall x P(x, y)$ $\exists e, 3, 4-7$
9. $\exists y \forall x P(x, y)$ $\forall e, 1, 2-7$



1. $\exists y \forall x P(x, y)$ premise
2. y_0 fresh variable
3. $\forall x P(x, y_0)$ assumption
4. x_0 fresh variable
5. $P(x_0, y_0)$ $\forall e, 3$
6. $\exists y P(x_0, y)$ $\exists i, 5$
7. $\forall x \exists y P(x, y)$ $\forall i, 4-6$
8. $\forall x \exists y P(x, y)$ $\exists e, 1, 2-7$



1. $\exists y \forall x P(x, y)$ premise
2. x_0 fresh variable
3. y_0 fresh variable
4. $\forall x P(x, y_0)$ assumption
5. $P(x_0, y_0)$ $\forall e, 4$
6. $\exists y P(x_0, y)$ $\exists i, 5$
7. $\exists y P(x_0, y)$ $\exists e, 1, 3-6$
8. $\forall x \exists y P(x, y)$ $\forall i, 2-7$

1. $\exists y \forall x P(x, y)$ premise
2. y_0 fresh variable
3. $\forall x P(x, y_0)$ assumption
4. x_0 fresh variable
5. $P(x_0, y_0)$ $\forall e, 3$
6. $\exists y P(x_0, y)$ $\exists i, 5$
7. $\exists y P(x_0, y)$ $\exists e, 1, 2-6$
8. $\forall x \exists y P(x, y)$ $\forall i, 4-7$

Points: **10**

11. Multiple Choice: Natural Deduction (prop) (var):
Consider the following proof with Nat...

ooo

Question

Consider the following proof with Natural Deduction for Propositional Logic, where we omitted the names of the proof rules.

- | | |
|-----|--|
| 1. | $(p \vee q) \wedge (q \rightarrow r)$ |
| 2. | $(q \rightarrow r)$ |
| 3. | $(p \vee q)$ |
| 4. | p |
| 5. | $r \vee p$ |
| 6. | q |
| 7. | r |
| 8. | $r \vee p$ |
| 9. | $r \vee p$ |
| 10. | $(p \vee q) \wedge (q \rightarrow r) \rightarrow (r \vee p)$ |

Which one of the following proof rules is NOT used in the proof above?

Answer

☒ introduction of conjunction

elimination of conjunction (left/right)

introduction of disjunction (left/right)

elimination of disjunction



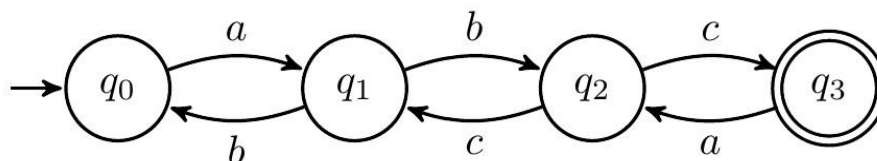
Points: **10**

12. Multiple Choice: NFA to DFA: Consider the following NFA M. Transfo...

ooo

Question

Consider the following NFA M. Transform it to a minimal, deterministic and complete DFA M'. How many states does M' have?



Answer

4 states

5 states

☒ 6 states

7 states

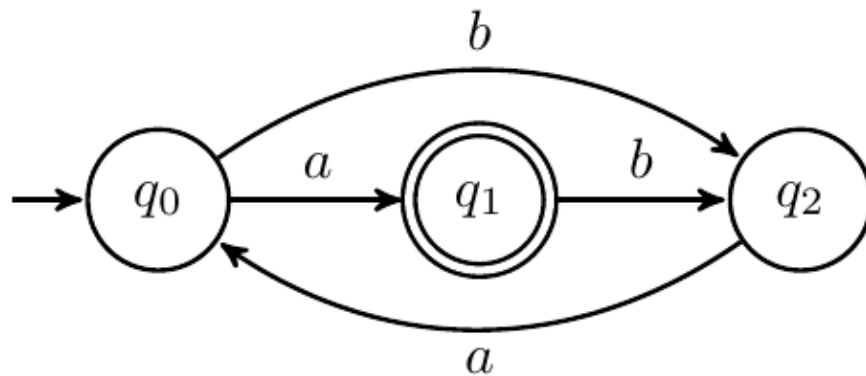
Points: **10**

13. Multiple Choice: DFA to RE (var): Consider the following (incomplete) D...

ooo

Question

Consider the following (incomplete) DFA.



This DFA accepts some language L. Which of the following regular expressions describes the same language L?

Answer
☒ $(ba)^* a (ba(ba)^* a)^*$
☐ $(ba)^* a (baa)^*$
☐ $(a + b) (baa)^*$
☐ $((ab + b) a)^*$
Points: **10**

14. Multiple Choice: PCP: Consider the following instance of Po...

ooo

Question

Consider the following instance of Post's correspondence problem, which consists of three tiles over the alphabet $\{a,b,c,d\}$:

$$\begin{pmatrix} ab \\ abc \end{pmatrix}, \begin{pmatrix} bcbd \\ d \end{pmatrix}, \begin{pmatrix} cb \\ bcb \end{pmatrix}$$

Answer

This PCP instance can be solved, even without repeating any of the tiles

✓ This PCP instance can be solved, but at least one tile must be repeated

This PCP instance has no solution

We cannot know if this PCP instance has a solution, since PCP is undecidable



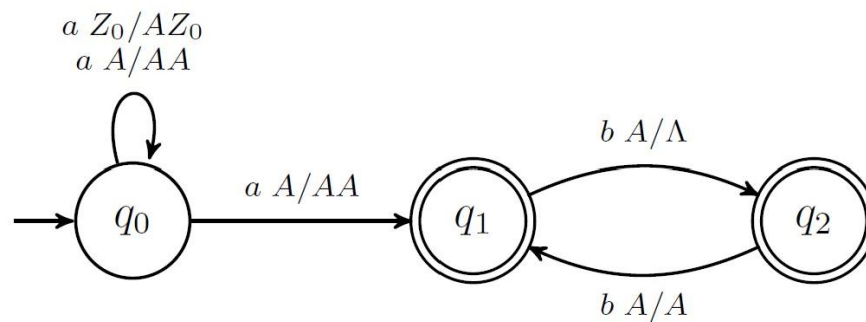
Points: **10**

15. Multiple Choice: PDA recognize (var): Consider the following PDA , with ini...

ooo

Question

Consider the following PDA M , with initial stack symbol Z_0 . We consider $\mathcal{L}(M)$, the language of M by final state.



Question: Which one of the following languages is $\mathcal{L}(M)$?

Answer

$\{ a^p b^q \mid p \geq 1 \text{ and } 2p \geq q \}$

$\{ a^p b^q \mid p \geq 2 \text{ and } 2p \geq q \}$

$\{ a^p b^q \mid p \geq 1 \text{ and } 2p > q \}$

✓ $\{ a^p b^q \mid p \geq 2 \text{ and } 2p > q \}$

Points: **10**

16. Multiple Choice: Problem reduction: Recall the following problems on Turi...

ooo

Question

Recall the following problems on Turing Machines:

- Accept- Λ : input a TM M , decide if M accepts the empty word Λ
- Subset: input two TMs, M_1 and M_2 , decide if $\mathcal{L}(M_1) \subseteq \mathcal{L}(M_2)$

We introduce some auxiliary Turing Machines that might be helpful:

- T_1 : a Turing Machine that accepts the language $\{\Lambda\}$, and leaves the tape unchanged
- T_2 : a Turing Machine that accepts the language Σ^* , and leaves the tape unchanged
- T_3 : a Turing Machine that erases all its input, and terminates successfully

We write $T_1 \rightarrow T_2$ for the sequential composition of T_1 and T_2 .

We will also write $\langle T_1, T_2 \rangle$ for the pair of inputs T_1 and T_2 .

Question: Which of the following functions is an appropriate reduction function, to show that $\text{Accept} - \Lambda \leq \text{Subset}$?

Answer

✓ $F(M) := \langle T_2, T_3 \rightarrow M \rangle$

$F(M) := \langle T_2, T_1 \rightarrow M \rangle$

$F(M_1, M_2) = \overline{M_1} \cap M_2$

(i.e., a TM that simulates the product of M_2 and the complement of M_1)

Points: **10**

17. Multiple Choice: prove not context-free: What is the most reasonable method to...

ooo

Question	What is the most reasonable method to prove that a language L is not context-free?
Answer	Prove that Post's Correspondence Problem can be reduced to L Show a non-terminating Turing Machine that accepts L ☒ Apply the Pumping Lemma Apply Rice's Theorem

Points: **10**

18. Multiple Choice: Regular Language (var): We consider languages over the alphab...

ooo

Question	We consider languages over the alphabet $\{a,b,c\}$. Which one of the languages below is NOT a regular language?
Answer	☒ The set of words where the difference of the number of a's and b's is greater than 2 The set of words where the sum of the number of a's and b's is greater than 2 The set of words where the sum of the number of a's and b's is even The set of words where the difference of the number of a's and b's is even

Points: **10**

19. Multiple Choice: Rice's Theorem Condition (var): Rice's Theorem states that under cert...

ooo

Question

Rice's Theorem states that under certain conditions, a decision problem on Turing Machines is undecidable. The problems below all take as input a single Turing Machine M on alphabet $\{0,1\}$. Which one of these problems satisfies the conditions of Rice's Theorem?

Answer

$\mathcal{L}(M) \subseteq \{0,1\}^*$

The number of states in M is a square number

✓ M accepts the word "0110"

On input "0110", M will write two consecutive 0's on the tape



Points: **10**

20. Multiple Choice: satisfiability propositional logic: Which one of the following formulas o...

ooo

Question	Which one of the following formulas of propositional logic is satisfiable, but not valid?
Answer	$(a \vee b) \wedge \neg a \wedge \neg c \wedge (b \rightarrow c)$ $((a \vee b) \wedge \neg a \wedge \neg c) \rightarrow b$ ✓ $(a \vee b) \wedge \neg a \wedge (\neg c \rightarrow b)$ $((a \vee b) \wedge \neg a) \rightarrow (\neg c \vee b)$



Points: **10**

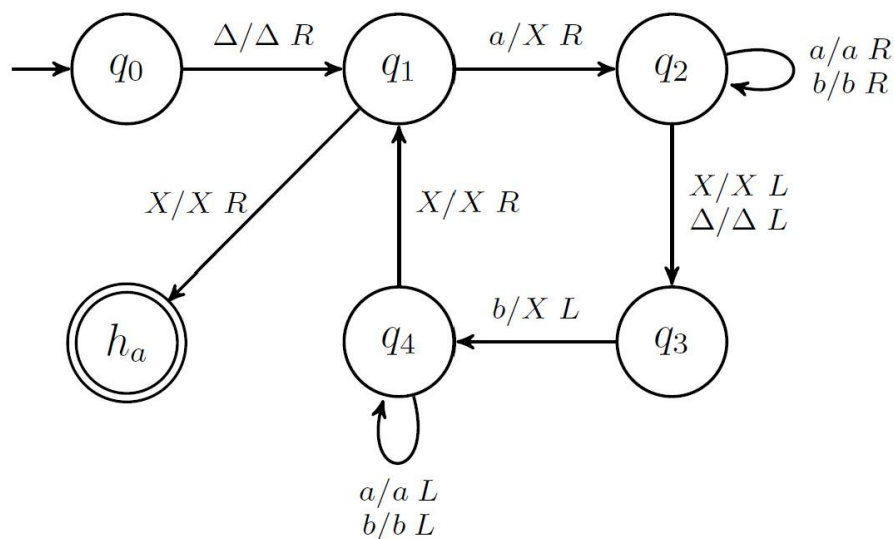
21. Multiple Choice: TM recognize (var): Consider the following Turing Machine...

ooo

Question

Consider the following Turing Machine, with input alphabet $\{a, b\}$ and tape alphabet $\{a, b, \Delta\}$.

Transitions to the rejecting state h_r are omitted from the diagram.



Which one statement about this TM is true?

Answer

This TM accepts the language $\{a^k b^k \mid k > 0\}$, but does not decide it.

✓ This TM decides the language $\{a^k b^k \mid k > 0\}$.

This TM accepts the language $a^* b^*$, but does not decide it.

This TM decides the language $a^* b^*$.



Points: **10**

22. Multiple Answer: True or False (var): Select all correct statements below

ooo

Question

Select all correct statements below

Answer

✓ Every NFA can be transformed to an equivalent DFA (deterministic FA)

Every (non-deterministic) PDA can be transformed to an equivalent DPDA (deterministic PDA)

- ✓ Every NTM can be transformed to an equivalent (deterministic) TM
- ✓ Every (non-deterministic) PDA can be transformed to a (deterministic) TM

Points: **10**

23. Multiple Choice: True or False (var): Which one of the following statements...

ooo

Question	Which one of the following statements (about languages over some finite alphabet) is WRONG?
Answer	Every context-free language is recursive. Every regular language is deterministic context-free. ✓ Every recursive language is context free. Every context-free language is countable.

Points: **10**

24. Multiple Choice: Validity and Provability: There exists a formula P that is:

ooo

Question	There exists a formula P that is:
Answer	✓ valid in arithmetic, but not provable in Peano Arithmetic provable in Peano Arithmetic, but not valid in arithmetic valid in predicate logic, but not provable with natural deduction provable with natural deduction, but not valid in predicate logic

Points: **10**

25. Multiple Choice: Validity and Satisfiability: Which of the following statements is ...

ooo

Question	Which of the following statements is WRONG?
Answer	☒ $\forall x \forall y (x + y = y + x)$ is valid in predicate logic ☐ $\forall x \forall y (x + y = y + x)$ is satisfiable in predicate logic ☐ $\forall x \forall y (x + y = y + x)$ is valid in arithmetic ☐ $\forall x \forall y (x + y = y + x)$ is satisfiable in arithmetic

Points: **10**

26. Multiple Choice: Which is context-free (var): Which one of the following languages ...

ooo

Question	Which one of the following languages is context-free? We consider only words over the alphabet $\{a, b, c\}$, and $n_a(w)$ denotes the number of a's that occur in the word w, and i and j are natural numbers.
Answer	☐ $a^i b^{2i} c^i$ ☒ $a^i b^{i+j} c^j$ ☐ $\{ w \mid n_a(w) = n_b(w) \text{ and } n_a(w) = n_c(w) \}$ ☐ $a^i b^j c^i b^j$

Points: **10**

27. Multiple Answer: Chomsky Hierarchy (var): Which of the following statements are...

ooo

Question

Which of the following statements are correct? (you have to select all correct answers)	
Answer	☒ Every Regular Grammar can be transformed to an equivalent Deterministic Finite Automaton
	☒ Every Regular Expression can be transformed to an equivalent Context-Free Grammar
	☒ Every Deterministic Push-Down Automaton can be transformed to an equivalent Turing Machine
	Every Terminating Turing Machine can be transformed to an equivalent Push-Down Automaton
	Every Unrestricted Grammar can be transformed to an equivalent Context-Free Grammar

Points: **10**

28. Multiple Choice: propositional logic equivalence (var):
All formulas in propositional logic b...

ooo

Question	All formulas in propositional logic below are equivalent, except one.
	Which formula is NOT equivalent to the others?
Answer	$(a \vee \neg b) \wedge (c \vee a)$
	$a \vee (\neg b \wedge c)$
	$(\neg a) \rightarrow (\neg b \wedge c)$
	☒ $(\neg b \wedge c) \rightarrow a$

Points: **10**

29. Multiple Answer: grammar syntax (var): Consider the
following grammar G, wit...

ooo

Question

Consider the following grammar G, with terminals {a,b} and non-terminals {S,A,B} (S is the start symbol):

$$S \rightarrow b A \mid B$$

$$A \rightarrow A B \mid B$$

$$B \rightarrow a \mid \Lambda$$

Select all correct statements on grammar G below

Answer

☒ The start symbol S is nullable

☐ G has no chain rules

☐ G is in Chomsky Normal Form

☒ G is a context-free grammar

☐ G is a regular grammar

☐ Each word in $L(G)$ has at most one terminal symbol "a"



Points: **10**

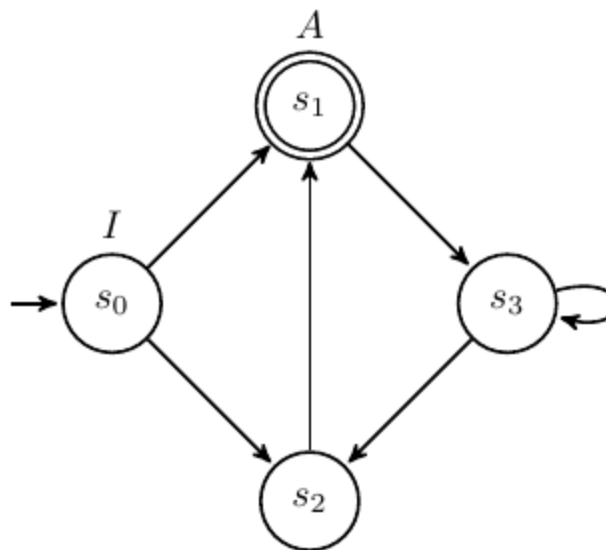
30. Multiple Answer: 1st-order model: Consider the following model of predi...

ooo

Question

Consider the following model of predicate logic, with relation symbols $\{I, A, R\}$.

In the model, " $I(x)$ " is interpreted as " x is the initial state", " $A(x)$ " as an " x is an accepting state", and " $R(x,y)$ " as "There is a transition from x to y ".



Below are a number of formulas in predicate logic. Select all formulas that hold in (are satisfied by) the model above.

Answer

☒ $\exists x R(x,x)$

$\forall x (\exists y (R(x,y) \wedge A(y)))$

☒ $\forall x \forall y ((A(x) \wedge A(y)) \rightarrow x = y)$

☒ $\exists x \exists y \exists z (R(x,y) \wedge R(y,z) \wedge R(z,x))$

$\exists x (A(x) \wedge I(x))$



Points: **10**

31. Multiple Choice: valid/sat in propositional logic (var):
Consider the following formula in pro...

ooo

Question

Consider the following formula in propositional logic:

$(q \rightarrow p) \vee (p \rightarrow q)$. What property does it have?

Answer



It is both satisfiable and valid

It is satisfiable, but not valid

It is valid, but not satisfiable

It is neither valid, nor satisfiable

Points: **10**



32. Multiple Choice: Ambiguity: Below are four grammars, where S and ...

ooo

Question	Below are four grammars, where S and T are non-terminals (S is the start symbol), and o,+, (,) are terminal symbols. Which one of these four grammars is ambiguous?
Answer	✔ $S \rightarrow S + S \mid \emptyset$ $S \rightarrow (S + S) \mid \emptyset$ $\begin{cases} S \rightarrow S + T \mid \emptyset \\ T \rightarrow \emptyset \end{cases}$ $\begin{cases} S \rightarrow T + S \mid T \\ T \rightarrow \emptyset \end{cases}$

Points: **10**



33. Multiple Choice: Free and Bound variables (var):
Consider the following formula in pre...

ooo

Question	Consider the following formula in predicate logic: $\forall z (P(z, x) \rightarrow \exists x Q(z, x))$ Which variables occur bound and free in this formula?
Answer	x and z occur free, and only z occurs bound

- ☒ only x occurs free, and x and z occur bound
- ☐ x and z occur bound, and the formula has no free variables
- ☐ x and z occur free, and the formula has no bound variables
- ☐ only x occurs free, and only z occurs bound

Points: **10**

34. Multiple Answer: Overlap in PDA (var): In a Deterministic PDA, two transitio...

ooo

Question

In a Deterministic PDA, two transitions from the same state should not be overlapping. Below, we show the labels of a number of transitions from the same state of some PDA. You may assume that X, Y and A are different stack symbols, and a and b are different alphabet symbols.

Select all pairs of overlapping transitions below

Answer

☒ $\Lambda X/A$ and $a X/A$

☐ $\Lambda X/A$ and $\Lambda Y/A$

☒ $a X/A$ and $a X/Y$

☐ $a X/A$ and $a Y/A$

☐ $a X/A$ and $b X/A$

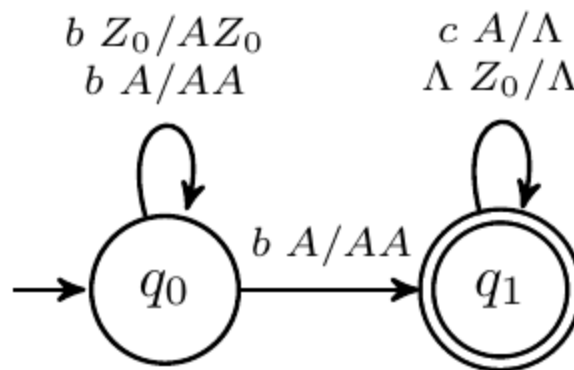
Points: **10**

35. Multiple Choice: PDA acceptance: Consider the following Push-Down Auto...

ooo

Question

Consider the following Push-Down Automaton (PDA), called M:



We write $L_E(M)$ for the language that M accepts with empty stack.

We write $L_A(M)$ for the language that M accepts with accepting state.

What are these languages?

Answer

$$L_A(M) = \{ b^m c^n \mid n = m \wedge m > 1 \}$$

$$L_E(M) = \{ b^m c^n \mid n \leq m \wedge m > 1 \}$$

✓ $L_A(M) = \{ b^m c^n \mid n \leq m \wedge m > 1 \}$

$$L_E(M) = \{ b^m c^n \mid n = m \wedge m > 1 \}$$

$$L_A(M) = \{ b^m c^n \mid n = m \wedge m \geq 1 \}$$

$$L_E(M) = \{ b^m c^n \mid n \leq m \wedge m \geq 1 \}$$

$$L_A(M) = \{ b^m c^n \mid n \leq m \wedge m \geq 1 \}$$

$$L_E(M) = \{ b^m c^n \mid n = m \wedge m \geq 1 \}$$



Points: **10**

36. Multiple Answer: Recursively Enumerable (var): Assume that the following statement h...

ooo

Question

Assume that the following statement holds:

"L is a recursively enumerable language"

Select all statements below that follow from this statement.

Answer

- ☒ There exists a TM that accepts precisely all words w in L
- ☒ There exists a TM that accepts all words w in L and hangs on all words w not in L
- ☐ There exists a TM that accepts all words in L and rejects all words not in L
- ☐ There exists a TM that decides the language L
- ☒ There exists a TM that enumerates all words in L



Points: **10**

37. Multiple Answer: Regular Expressions (var): Consider the following regular expres...

ooo

Question

Consider the following regular expression:

$((ab)^* + ac)^*$

Select all regular expressions below that describe the same language:

Answer

$(ab)^* + (ac)^*$

☒ $(ab + ac)^*$

☒ $(a(b + c))^*$

$$a * (b + c) *$$

Points: **10**

38. Multiple Choice: Truth table (var): What is the correct truth table of th...

ooo

Question

What is the correct truth table of the formula

$$(p \rightarrow \neg q) \rightarrow p$$

Answer

✔

p	q	$(p \rightarrow \neg q) \rightarrow p$
T	T	T
T	F	T
F	T	F
F	F	F

p	q	$(p \rightarrow \neg q) \rightarrow p$
T	T	T
T	F	T
F	T	F
F	F	T

p	q	$(p \rightarrow \neg q) \rightarrow p$
T	T	T
T	F	T
F	T	T
F	F	T

p	q	$(p \rightarrow \neg q) \rightarrow p$
T	T	F
T	F	T
F	T	F
F	F	F

Points: **10**

39. Multiple Choice: DFA minimal (var): Which of the following DFAs is minima...

ooo

Question	Which of the following DFAs is minimal?
Answer	Only M_3 minimal Only M_1 is minimal M_2 and M_3 are minimal but M_1 is not minimal ✓ Only M_2 is minimal

Points: **10**

40. Multiple Choice: Decision problems on CFG (var): Which of the following problems on Co...

ooo

Question	Which of the following problems on Context-Free Grammars is undecidable? (here $L(G)$ is the set of words generated from the start symbol of G)
Answer	Given a CFG G and a word w, decide if w is in $L(G)$ Given a CFG G, decide if there exists any word in $L(G)$ Given two CFGs G_1 and G_2, and a word w, decide if w is contained in both $L(G_1)$ and $L(G_2)$ ✓ Given two CFGs G_1 and G_2, check if there is any word in both $L(G_1)$ and $L(G_2)$